

S.no	Research Paper title	authors	year	Problem addressed	Proposed methodology	technology used	key results	research gaps
1	Advanced System for Remote Updates on ESP32-Based Devices Using Over-the-Air Update Technology	Lukas Formanek	2025	Manual firmware updates are costly, insecure, and impractical for large-scale heterogeneous IoT deployments	Proposed centralized, multi-protocol OTA architecture with version control, rollback, and performance analysis	ESP32, Wi-Fi, BLE, Zigbee, LoRa, GSM	Reduced update latency, reliable large-scale updates, scalable OTA management	Security analysis and interoperability beyond ESP32 not fully explored
2	Holistic IoT Architecture for Secure Lightweight Communication, Firmware Update, and Trust Monitoring	Jesus Sanchez-Gomez	2021	Lack of interoperable and trustworthy security management for large, low-power IoT networks	Designed holistic architecture combining trust monitoring and firmware updates using open standards	LoRaWAN, LO-GAAP-EAP, SCHC, OSCORE, IPFS, Hyperledger	Secure trust monitoring and firmware distribution for constrained devices	Real-world deployment scalability and performance evaluation limited
3	A Lightweight, End-to-End Encrypted Data Pipeline for IIoT: An AES-GCM Implementation for ESP32, MQTT, and Raspberry Pi	Gulshat Amirkhanova	2026	TLS-to-broker MQTT model fails to protect data from compromised brokers	Implemented application-layer end-to-end encryption using AES-GCM	ESP32, MQTT, Raspberry Pi, AES-256-GCM, mbedTLS	99.72% success rate, strong integrity and confidentiality, acceptable latency	Key management and large-scale deployment automation not addressed
4	OTA: An Approach to Secure Over-the-Air Updates on the Internet of Things Scenario	Cleber S. Peter	2021	OTA updates are vulnerable to security attacks in distributed IoT systems	Proposed OTA approach aligned with IETF SUIT security requirements	SUIT specification	Improved protection against common OTA attacks	Only partial evaluation; needs full validation and implementation
5	Secure End-to-End Communications with Lightweight, Cryptographic Algorithm	Augustine Ukpebor	2018	Traditional cryptography is resource-heavy for constrained IoT devices	Implemented and benchmarked lightweight KATAN32 cipher on ESP32	ESP32, KATAN32, Wi-Fi, Base64, JSON	Lower memory, power usage, and comparable throughput to AES	Automatic server-side decryption and scalability not explored
6	Design and Implementation of MQTT-Based Over the Air, Updating Against Curious Brokers	Nian-Zu Wang, Hung-Yu Chien	2023	MQTT-based OTA assumes broker trust, risking privacy	Designed E2E-secured MQTT OTA scheme with formal verification	MQTT, E2E channel, AVISPA	Strong privacy against curious brokers with minimal overhead	Integration with existing commercial IoT platforms not evaluated
7	MUP: Simplifying Secure Over-the-Air Update with MQTT for Constrained IoT Devices	Kristina Sahlmann	2021	Secure and fresh firmware updates are difficult for constrained MQTT devices	Proposed MUP protocol with versioning, nonces, and two-phase updates	MQTT, Contiki-NG, CC2538, SHA-256, ECDSA	Authenticated, replay-safe OTA with reduced resource usage	Weak bootstrap key distribution, limited crypto agility
8	A Novel MQTT 5.0-Based Over-the-Air Updating Architecture Facilitating Stronger Security	Hung-Yu Chien, Nian-Zu Wang	2022	Existing MQTT OTA models expose data to third-party brokers	Designed MQTT 5.0 OTA architecture with inner E2E secure channel	MQTT 5.0, TLS, EC0H, ECDSA	Enhanced privacy, fewer MQTT interactions, broker-independent security	Outer-channel encryption overhead and cross-application use
9	Implementation of SSL/TLS Security with MQTT Protocol in IoT Environment	Iqbal Luqman	2023	SSL/TLS causes high energy and overhead on IoT devices	Implemented and analyzed TLS-enabled MQTT under stress conditions	SSL/TLS, MQTT, ESP32, Raspberry Pi	TLS consumes 4x more energy, impractical for battery IoT	Need low-power security alternatives for IoT
10	LightQSign: Lightweight and Quantum-Safe Signatures for Heterogeneous IoT Applications	Attila A. Yavuz	2023	Traditional and PQ signatures are too heavy for IoT	Proposed LightQSign lightweight quantum-safe signature	LightQSign, hash-based crypto, A18Mega128A1	56–76x better energy efficiency, compact signatures	Real-world IoT OTA and authentication integration not evaluated
11	Efficiency and Security Evaluation of Lightweight Cryptographic Algorithms for Resource-Constrained IoT Devices	Indu Radhakrishnan, Shruti Jadau, Prasad B. Honnavalli	2024	Traditional cryptographic algorithms are resource-intensive for constrained IoT devices, making security-performance balance difficult	Implemented and evaluated AES-128, SPECK, and ASCON on Arduino boards using execution time, memory, latency, throughput, and security analysis	AES-128, SPECK, ASCON, Arduino Nano, Arduino Micro, C, Arduino IDE	AES fastest, SPECK lowest latency and highest throughput, ASCON lowest RAM usage and strong security	Trade-offs between security and performance not deeply analyzed; limited to specific Arduino platforms
12	Review and Analysis of FPGA and ASIC Implementations of NIST Lightweight Cryptography Finalists	Evangelia Konstantopoulou, George Athanasios, Nicolas Sklavos	2025	Difficulty in selecting suitable NIST lightweight cryptography algorithms for IoT due to diverse hardware constraints	Reviewed and compared FPGA and ASIC implementations of NIST LWC finalists with optimized and baseline designs	ASCON, TinyJAMBU, Grain128-AEAD, FPGA, ASIC	ASCON most balanced, TinyJAMBU and Grain128-AEAD suitable for highly constrained and low-latency environments	Lack of real-world IoT deployment validation; comparison limited to existing studies
13	Design and Implementation of a Secure Communication Architecture for IIoT Devices	Cezar-Gabriel Dumitrache, Petre Angelescu	2025	WP2A-personal security flaws make IIoT wireless communication vulnerable	Designed and implemented 802.1X-based secure architecture using certificate-based authentication	Raspberry Pi, 802.1X, EAP-TLS, EAP-MSCHAPv2, digital certificates	Improved security and energy efficiency; scalable secure wireless communication	Limited evaluation on highly resource-constrained IIoT devices; lack of comparison with alternative architectures
14	Automatic Fingerprinting of Vulnerable BLE IoT Devices with Static UUIDs from Mobile Apps	Chaoshun Zuo, Haoshuang Wen, Zhiqiang Lin, Yinduan Zhang	2019	BLE IoT devices can be fingerprinted and attacked due to static UUIDs, insecure "Just Works" pairing, and weak app-level authentication	Identified BLE vulnerabilities, developed BLEScope tool to analyze mobile apps, and conducted large-scale field testing with BLE sniffing	BLE, Android apps, BLEScopeE, binary analysis, Raspberry Pi, BLE antenna	94.6% of detected BLE devices fingerprintable; 7.4% vulnerable to unauthorized access	Need for dynamic UUID adoption and stronger app-level authentication
15	Protocols for Checking Compromised Credentials	Lucy Li, Bijeeta Pal, Junade Ali, Nick Sullivan, Rahul Chatterjee, Thomas Ristenpart	2019	Existing compromised credential checking services leak sensitive information via hash-prefix techniques	Formal security analysis, leakage measurement, and design of privacy-preserving protocols (FSB, IDB)	C3 services, hash functions, PSI, OPRF, FSB, IDB	Hash-prefix leakage enables up to 12x stronger guessing attacks; FSB and IDB reduce or eliminate leakage	PSI scalability limits; IDB may expose usernames
16	MQTree: Secure OTA Protocol Using MQTT and MerkleTree Embedded IoT Systems	Yunye Shin, Sanghoon Jeon	2024	Firmware integrity and efficiency issues in automotive OTA updates using HTTPS or blockchain-based methods	Proposed MQTree combining MQTT (TLS) with Merkle-tree-based integrity verification in centralized OTA	MQTT, TLS, Merkle Tree, SHA-256, HTTPS, HSM, Jetson Orin, Arduino MCU	4x faster than HTTPS; strong integrity verification even if TLS is breached	Limited real-world validation for automotive OTA environments
17	Lightweight and Resilient Signatures for Cloud-Assisted, Embedded IoT Systems	Saif E. Nouna, Attila A. Yavuz	2023	Lightweight, scalable, and compromise-resilient authentication is difficult for cloud-assisted IoT devices	Designed LRSHA and FLRSHA signatures with commitment separation and TEE-assisted distributed verification	Digital signatures, TEEs, PRF, hash functions, EC crypto, 8-bit AVR MCU	Up to 46x faster than Ed25519; compact signatures; strong forward security	Large-scale IoT-to-cloud deployment validation needed
18	Efficient Implementation of CRYSTALS-KYBER Key Encapsulation Mechanism on ESP32	Fabian Segatz, Muhammad Ihsan Al Haiz	2022	Implementing post-quantum Kyber KEM efficiently on resource-constrained ESP32 IoT devices	Dual-core task partitioning, use of hardware accelerators, empirical cycle-count analysis using ESP-IDF	CRYSTALS-Kyber, ESP32, FreeRTOS, ESP-IDF, MbedTLS	Up to 1.84x speed-up using dual-core + accelerators; Kyber512-90s most efficient	Further parallelization (matrix A generation); optimized task scheduling; lack of comparative ESP32 benchmarks
19	Bibliometric Analysis of IIoT Lightweight Cryptography	Zenith Dewamani, Bharandharan Shanmugam, Sami Azam, Suresh Thennadil	2023	Need to identify research trends and gaps in IIoT lightweight cryptography	Bibliometric performance and science mapping of 2015–2023 publications	R, VOSviewer, Bibliometrix, Scopus, WOS, IEEE Xplore	India leads research; IEEE Access most influential; LWC a growing field	Need finer-grained studies on specific ciphers and cryptographic primitives
20	RAM-MEN: Robust Authentication Mechanism for IIoT-Enabled, Edge Networks	Muhammad Tanveer, Saud Alhagaj, Adnan Aljiz	2024	Existing authentication mechanisms in MEC-IIoT are vulnerable to impersonation and insider attacks	Proposed RAM-MEN protocol; formal verification (Scyther, ROR); performance evaluation	PIUFs, ECC, ASCON, SHA-256, Raspberry Pi, MIRACL	21–45% lower communication cost; 17–83% lower computation cost; strong attack resistance	Large-scale real-world deployment and integration with 6G MEC networks
21	A Lightweight Image Cryptosystem for Cloud-Assisted Internet of Things	Esau Talwo Oladipupo, Oluwakemi Christiana Abokoye, Joseph Bamidele Awotunde	2024	Multimedia data security in IIoT is difficult with S-box-based lightweight cryptosystems	Proposed HXP algorithm using XOR, permutation, and hashing; extensive statistical analysis	HXP algorithm, SHA-256, XOR, bit permutation	Eliminates S-box; high NPCR/UACI; strong resistance to statistical attacks	Hardware-level evaluation and real-time IIoT deployment not explored
22	Guidance Framework for Developing IIoT-Enabled Systems' Cybersecurity	Hezam Akram Abdulghani, Anastasija Colfen, Niels Alexander Nijdam	2023	Lack of unified cybersecurity frameworks for heterogeneous IoT systems	Proposed SLC-based security framework with classification and mitigation phases	Security Level Certificates, IIoT asset classification	Structured, scalable IIoT cybersecurity guidance	Practical implementation and quantitative validation needed
23	uJUF: A Framework for Detecting Over-the-Air Malicious Updates in Autonomous Vehicles	Anam Qureshi, Murk Marvi, Jawwad Ahmed Shamsi, Adrian Aljiz	2021	OTA updates in autonomous vehicles vulnerable to malicious software injection	Enhanced Uptane with CNN-based malware detection	CNN, Deep Learning, Uptane, Windows/Linux executables	Up to 97.5% detection accuracy; ~3 ms detection time	Transfer learning and Linux-specific detection accuracy improvements
24	An Efficient Certificateless Forward-Secure Signature Scheme for Secure IIoT Deployments	Tahir Ali Shah, Insaif Ullah, Muhammad Asghar Khan, Pascal Lorenz, Nisreen Imab	2023	High cost and key escrow issues in IIoT digital signature schemes	Proposed HECC-based certificateless forward-secure signature with ROM analysis	HECC, SHA hashes, ROM, 5G, Bluetooth	Up to 97% lower computation cost; reduced communication overhead	Implementation on real IIoT hardware not evaluated
25	A Decentralized Trust Establishment Protocol for Smart IIoT Systems	Driss El Majdoubi, Hanan El Bakkali, Mohamed Bensaih, Souad Sadki	2022	Centralized trust models unsuitable for resource-constrained IIoT	Blockchain-based SAS authentication with ECC and decentralized identity	Blockchain, ECC, SAS, EC0H, ECIES	<90 ms authentication; strong resistance to major attacks	Blockchain overhead and scalability in large IIoT deployments
26	Over-the-Air Upgrading for Enhancing Security of Intelligent Connected Vehicles: A Survey	Beibei Li	2024	Lack of comprehensive understanding of OTA security risks in ICVs	Systematic survey of OTA threats, techniques, and mitigations	OTA (SOTA/OTA), IDS, TPM, Secure Boot, TLS	Identifies OTA risks and mitigation strategies; highlights OTA importance	End-to-end cloud–network–vehicle security still unresolved
27	Federated Learning for IIoT Devices: Enhancing TinyML with On-Board Training	M. Ficco	2024	TinyML lacks on-device training while ensuring privacy and efficiency	Combined FL and TL; evaluated on heterogeneous IIoT hardware	Federated Learning, Training Learning, TinyML, MQTT, Genann	Comparable accuracy to centralized training; ESP32 lowest power use	Further accuracy improvement and adaptive data sampling
28	A Zero Trust and Blockchain-Based Defense Model for Smart Electric Vehicle Chargers	Peirong Li, Wei Ou, Haozhe Liang, Wenbao Han, Qiongli Zhang, Guang Zeng	2023	Multiple cybersecurity threats in smart EV chargers and charging cloud platform	Zero Trust IAM, blockchain-based key management, SM cryptography, secure OTA with threshold	Hyperledger Fabric, SM2/SM3/SM4, SMSSL, Redis	Secure communication, anti-replay and anti-tampering; meets national standards	Broader IoEV ecosystem validation and large-scale deployment analysis
29	Overcoming the DDoS Attack Vulnerability of an ISO 15847 Shipboard Data Server	Changliu Lee, Sejong Lee	2023	MQTT-based shipboard data servers vulnerable to DDoS attacks	Modified MQTT v5 with minimum publish interval enforced by broker	MQTT v5, Mosquitto, Raspberry Pi, L4 switches	~10x reduction in CPU, memory, and network usage during attacks	Formal standardization and wider maritime deployment evaluation
30	Optimizing Electric Racing Car Performance through Telemetry-Integrated Battery Charging	A. F. Villa-Salazar	2024	Inefficient battery charging increases time and thermal stress in electric racing cars	Telemetry-based data collection and Response Surface Methodology optimization	ESP32, LoRa, Arduino Mega, Raspberry Pi, RStudio	Optimal current (2.38 A) minimized charging time; trade-off with temperature identified	Integrating temperature as a controllable optimization variable
31	A Tailored Discourse on Microcontrollers, Single Board Computers and Their Applications to Monitor and Control Chemical Reactions	Daniel E. Fitzpatrick, Matthew O'Brien, Steven V. Ley	2019	Chemists lack practical knowledge of embedded systems for reaction monitoring	Educational discourse with hands-on programming and lab demonstrations	ESP32, Raspberry Pi, Arduino, Python, C/C++, SimpleCV	Enables chemists to adopt embedded systems for real-time control	No quantitative performance evaluation or advanced automation focus
32	Delay and Energy Consumption of MQTT over QUIC: An Empirical Characterization	Sidna Jeddou	2022	Uncertainty about QUIC suitability for IIoT regarding delay and energy use	Real-hardware testbed comparing MQTT over TCP/TLS vs QUIC	MQTT, QUIC, TCP, TLS, Raspberry Pi, Linux TC	QUIC reduced latency without increasing energy consumption	Performance under heavy congestion and higher traffic loads
33	Highly Accurate and Reliable Wireless Network Slicing in 5G Networks	Sulaiman Khan	2021	Inefficient slice allocation and load balancing in 5G/6G networks	Hybrid CNN–LSTM model for traffic classification and slice assignment	CNN, LSTM, TensorFlow, Keras, NS2	95.17% slice allocation accuracy; robust failure handling	Real-world deployment and mobility-aware enhancements
34	Inter-UAV Routing Scheme Testbeds	Georgios Amponis	2021	Lack of realistic hardware testbeds for UAV routing evaluation	Designed modular hardware testbed with multi-mode operation	Raspberry Pi Zero, ESP32, MPU6050, ESP-NOW, WiFi	Low-cost, flexible routing evaluation; <2.5 W power usage	Validation with real routing protocols and large UAV swarms
35	Securing MQTT by Blockchain-Based OTP Authentication	Francesco Buccafurri, Vincenzo De Angelis, Roberto Nardone	2020	MQTT lacks lightweight secure authentication mechanism	Blockchain-based OTP authentication using smart contracts	MQTT, Ethereum blockchain, smart contracts, SHA-256	Secure authentication without TLS; reduced energy overhead	Experimental validation and smart contract cost analysis
36	A Lightweight Hierarchical Method for Improving Security in IIoT Using Fuzzy Logic	Shadi Doostani, Hamid Barati, Ali Barati	2023	IIoT networks vulnerable to privacy and spoofing attacks	Fuzzy logic trust evaluation and lightweight hop-by-hop encryption	Fuzzy logic, column transposition cipher, NS-2	Improved packet delivery and spoofing resistance	Hardware implementation and real-time performance validation
37	Conv-Random Forest Based IoT for Valvular Heart Disease Classification	Tanmay Sinha Roy, Joyanta Kumar Roy, Nirupama Mandal	2023	Need for accurate real-time heart sound classification in remote healthcare	CNN feature extraction combined with Random Forest classification	ESP32, Raspberry Pi 4B, CNN, RF, MQTT, TensorFlow	~99.37% accuracy; low-cost portable diagnostic system	Larger clinical trials and reduced auscultation time
38	A Malicious Behavior Awareness and Defense Countermeasure Based on LoRaWAN Protocol	Shu-Yang Gao, Xiao-Hong Li, Mao-De Ma	2019	Replay and malicious attacks in LoRaWAN, especially under compromised root keys	Proposed SPT with OTP-based join-request redesign; 10mnet++	LoRaWAN v1.1, OTP, AES-128, BAN Logic, Scyther, 10mnet++	Defends replay attacks even with leaked root key; lightweight with acceptable power usage	Other attack vectors and long-term scalability not studied

	PKIdIoT: Towards Public Key Infrastructure for the Internet of Things	Joel Höglund, Samuel Lindemer, Martin Furuheid, Shahid Raza	2019	Lack of scalable PKI for resource-constrained IoT devices	Automated certificate enrollment with CBOR-based X.5PKI, X.509, CBOR, CoAP, DTLS, ECC, Contiki OS		>50% certificate size reduction; lower latency and energy use	DoS protection, revocation handling, and physical key compromise
39	Exploring the Potential of Blockchain Technology in an IoT-Enabled Environment: A Review	P Gulia, NS Gill, M Yahya, P Gupta, PK Shukla	2024	Security, privacy, and integrity challenges in IoT systems	Systematic literature review of blockchain-IoT integrations	Blockchain, Smart Contracts, Consensus Algorithms, Cryptography	Blockchain improves trust, integrity, and decentralization in IoT	Scalability, energy cost, and IoT-suitable consensus mechanisms
40	RT Box Card for Studying the Control Communication Impacts on Microgrid Performance and Stability	Maximiliano Aguilar	2022	Lack of real communication evaluation in microgrid simulations	Designed RT Box control card for real-protocol microgrids	ESP32, Raspberry Pi, RT Box, CAN, ESP-NOW, Bluetooth	Enables real-time protocol impact analysis on stability	Broader HIL integration and cybersecurity evaluation
41	A Secure and Collaborative Data Aggregation Scheme for Fine-Grained Data Distribution in IoT	Oladayo O. Olakanmi, Kehinde O. Odeyemi	2020	Security, privacy, and traffic overhead in heterogeneous IoT networks	Secure aggregation with ECC, perturbation, and mutual ECC, SHA-256, Bilinear Pairing, Arduino Uno		Low overhead, secure aggregation, resistance to major attacks	Cryptographic cost on very constrained devices
42	An IoT Architecture Based on MQTT and Node-RED for Monitoring Radon Gas	Alexandra Medina-Pérez	2021	Lack of low-cost real-time radon monitoring systems	MQTT-based IoT architecture with Node-RED and web MQTT, Node-RED, Raspberry Pi, ESP32, MySQL		Low-cost, real-time monitoring with minimal power usage	Alert automation and stronger data security needed
43	Distributed Ledger as a Service: A Web 3.0-Oriented Architecture	Francesco Chiti, Giorgio Gandini	2023	Secure, scalable M2M data storage without central authority	WoT architecture integrating MQTT with IOTA ledger	IOTA, MQTT, Fog Computing, OMNeT++	Improved latency, scalability, and data immutability	Dependence on evolving IOTA protocols and edge offloading
44	A Versatile and Customizable Low-Cost 3D-Printed Open Standard for Microscopic Imaging	Benedict Diederich	2020	High cost and inflexibility of commercial microscopy systems	Modular open-source 3D-printed microscopy platform	ESP32, Arduino, Raspberry Pi, MQTT, 3D Printing	€100–400 versatile imaging platform; reproducible designs	Material stability and camera sensitivity limitations
45	A Survey on Security Threats and Countermeasures in IoT Networks	Maria Papaioannou	2020	Security threats and privacy risks in IoT edge networks	Systematic categorization of threats and countermeasures	LoRa, ECC, ABE, DTLS, Lightweight Crypto	Comprehensive threat-countermeasure mapping	Lightweight authentication and access control remain open
46	Firmware Over-the-Air Programming Techniques for IoT Networks: A Survey	Konstantinos Arakadakis	2020	Complexity of secure and efficient OTAP in IoT networks	Survey of OTAP stages, techniques, and platforms	OTAP, Delta Updates, AES, RSA, Contiki, TinyOS	Structured OTAP framework and security insights	Energy efficiency, platform independence, secure group updates
47	Survey of Secure Routing Protocols for Wireless Ad Hoc Networks	Mehdi Boulaiche	2020	Traditional routing protocols in ad hoc networks lack security against malicious attacks	Systematic survey and classification of secure routing protocols, analysis of attacks and security requirements	Secure routing protocols (SEAD, S-DSR, SLSP, OLSR variants, AODV variants), cryptography, hash functions	Comprehensive taxonomy of secure routing approaches and attack mitigation techniques	Need for more robust, lightweight security for dynamic and resource-constrained networks
48	Secure Over-the-Air Software Updates in Connected Vehicles: A Survey	Subir Halder, Amrita Ghosal, Mauro Conti	2020	OTA updates in vehicles introduce serious security and privacy risks	Survey and comparative analysis of OTA approaches, threats, and evaluation setups	OTA updates, ECUs, cellular networks, in-vehicle networks	Identifies benefits, threats, and security requirements for vehicular OTA	Many open challenges in secure OTA design and deployment
49	An Ultra-Lightweight Mutual Authentication Protocol Based on LPN Problem with Distance Fraud Resistant	Kazem Mirzadi, Jamshid B. Mohasefi	2021	Existing RFID authentication protocols lack mutual authentication and are vulnerable to distance and MITM attacks	Designed lightweight mutual authentication and distance-bounding protocol, formal security proof	RFID, NFC, LPN problem, HB protocols, distance bounding, Toeplitz matrices	Resistant to mafia, terrorist, and distance fraud; post-quantum secure; lightweight for IoT	Practical deployment evaluation and performance benchmarking needed